

Global Dynamics of Connected Vehicle Systems

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Abstract

Road vehicles will vastly change in the coming years, due to the rise in autonomous vehicle technologies, sustainable forms of power, and better safety systems. This paper explores the improvements in traffic flow and safety that could be led by the shift to autonomously driven vehicles. Utilising VEHICLE-TO-VEHICLE (V2V) COMMUNICATION, it is possible to greatly reduce any time delays that arise due to human reaction times, or actuation delays. This model also takes into account any ACCELERATION SATURATIONS due to vehicle limitations. A model is developed that can work in an environment of both human-driven and connected autonomous vehicles (CAVs) - this lets us see how effective V2V communication is when in a HETEROGENEOUS environment, which represents the transition phase towards automated vehicle uptake (as automated vehicles are, at present, more expensive than ‘traditional’ vehicles). This model is computationally simulated in Python and, along with stability analysis, shows that the addition of autonomous vehicles has the ability to reduce, or even eliminate, traffic waves when in a heterogeneous environment. Additional nonlinearities are introduced into the system through the addition of obstacles, such as traffic lights and speed limits. We observe the effects of these obstacles on the heterogeneous system’s stability. The simulation is available as an open-source program that can be adapted to analyse other connected vehicle systems.

1 Introduction

1.1 Overview

This paper investigates whether V2V communications can improve both traffic flow and safety, by developing a model for the CONNECTED VEHICLE SYSTEM (CVS), followed by a CONNECTED CRUISE CONTROL (CCC) algorithm, that brings vehicles to their optimum velocities for a given HEADWAY between vehicles. We then investigate how the stability of the system is affected by several parameters, as well as the ratio of human-driven to automated vehicles. Some models are made for basic obstacles on the road, such as traffic lights and variable speed limits, and the effect on safety and the role of the automated vehicles are observed in this case. Individuals' differences, such as reaction times, and the roles of the length of the orange light for traffic lights, can be started to be investigated through a simulation in Python built on the model introduced. This paper then sees if it is possible to predict whether a connected vehicle system will be stable or not, using STABILITY ANALYSIS.

This model functions on the principle that no vehicles will use V2V communications to look ahead of the next CAV, meaning all vehicles will communicate to the CAV ahead of them, but not to vehicles ahead of that. The models used in this paper are presented by Kiss et al. in (1).

1.2 Key Equations and Background Information

N vehicles are placed on a loop of length L . There are M automated and therefore $N - M$ human-driven vehicles. Representing all vehicles in a set, each vehicle can be represented using indices. Human-driven vehicles are represented by $i_h \in H$, and autonomous vehicles $i_a \in A$. Therefore, $A + H = \{1, \dots, N\}$ and $|A| + |H| = N$. Following the models in (2), the function $f(a)$ can be produced, which is a smoothness function.

The value c is the smoothness range, limiting acceleration at a_{min} and a_{max} . As explained in the cited paper, this is necessary as a computational compensation.

Using this, the governing equations for both autonomous (Equation 1) and human-driven vehicles (Equation 2) can be given as:

where $i = i_a$:

$$\dot{v}_i = f(\alpha(V(h_i^\sigma) - v_i^\sigma) + \sum_{j=1}^{k_i} \beta_j(v_{i+j}^\sigma - v_i^\sigma)) \quad (1)$$

where $i = i_h$:

$$\dot{v}_i = f(\alpha_h(V(h_i^\tau) - v_i^\tau) + \beta_h(v_{i+1}^\tau - v_i^\tau)) \quad (2)$$

The input for the governing equation is h , standing for headway, which is defined as the distance from the front of the vehicle to the back of the next vehicle.

$V(h)$ is a curve representing the optimal velocity that is smooth at both h_{go} and h_{stop} and is a piecewise function as follows:

$$V(h) = \begin{cases} 0 & \text{if } h \leq h_{stop} \\ F(h) & \text{if } h_{stop} < h < h_{go} \\ v_{max} & \text{if } h \geq h_{go} \end{cases} \quad (3)$$

$$F(h) = \frac{v_{max}}{2} \left(1 - \cos \left(\pi \frac{h - h_{stop}}{h_{go} - h_{stop}} \right) \right) \quad (4)$$

This represents the optimal velocity that the vehicle would ideally go, and when passed through the function $f(a)$, will output an appropriate acceleration to take the vehicle to its optimum velocity. h_{go} and h_{stop} represent the headway at which the vehicle should drive at the speed limit or stop, respectively. Additionally, this OV curve is the same for both human and autonomous vehicles for the sake of simplicity. Future reports could take separate OV functions into account. Importantly, the velocity outputted by $V(h) \neq v_i$.

Note that $V(h_{stop}) = 0 \not\Rightarrow f(a) \leq 0$, as the velocity delta may cause the overall acceleration to be positive. Since autonomous vehicles take a sum of velocity deltas within their CASCADE (unlike humans that only look to the vehicle ahead) this effect is more prevalent with autonomous vehicles, and is a limitation of this model. This is further explained in the governing equation section of this report.

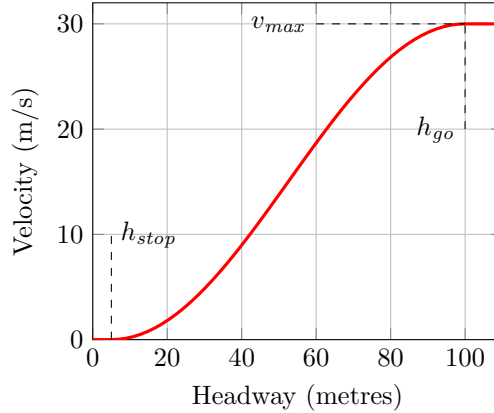


Figure 1: The optimal velocity (OV) function used in this paper, where $h_{stop} = 5$ metres, $h_{go} = 100$ metres and $v_{max} = 30$ m/s.

2 Governing Equations

2.1 Overview

We propose that to understand this equation intuitively, it is beneficial to break it into two parts, the VELOCITY ERROR and the VELOCITY DELTA.

The velocity error is given by the following:

$$\alpha(V(h_i^\theta) - v_i^\theta)$$

Note that $V(h)$ refers to the OV function, which gives the velocity the vehicle would ideally go at given the headway to the next vehicle (disregarding any delays or capabilities). The optimal velocity function used is from (1), which is based on real-world data. The methods used to collect the real-world data are explained in Appendix A.

In order to distinguish between moments in time before and after time delays, the notation v_i^θ is used to represent delays, where i is the index of the vehicle (within the set of $A + H$), and where θ is the time delay, such that $v_i^\theta = v_i(t - \theta)$, where t is the current time. Therefore the notation v_i^θ refers to the velocity of the vehicle with index i at a time that is θ before the current time (the time that the communication is sent from vehicle i is time $t - \theta$). This is necessary as due to time delays, the information that a human or autonomous vehicle uses to determine their current OV is the information from a time θ prior.

From this, we can see that the velocity error gives the *difference* between the ideal velocity and the velocity that the given vehicle i is going at now.

α (or α_h in the case of a human vehicle) has been described as a ‘sensitivity’ by Bando et al. in (3). This refers to the rate at which the vehicle will adjust to the OV, and is found by the reciprocal of the time it takes to adjust to the new velocity.

The velocity delta is given by the following:

where $i = i_a$:

$$\sum_{j=1}^{k_i} \beta_j (v_{i+j}^\sigma - v_i^\sigma)$$

The velocity delta part of the governing equation takes the sum of the differences in velocity between the autonomous vehicle i and all the human vehicles in front of it, which are indexed as $i + j$. Note that we do not look beyond the next autonomous vehicle. These are multiplied by the control parameter β_j . This parameter is directly related to the velocity delta and refers to the rate at which vehicle i will adjust to the sum of differences in velocity to the vehicles in front of itself. As the autonomous vehicle takes into account the velocities of all vehicles up to the next autonomous vehicle, this cascade of vehicle move smoothly as a group in the context of the entire system.

where $i = i_h$:

$$\beta_h(v_{i+1}^\tau - v_i^\tau)$$

We assume that human drivers only react to the velocity of the vehicle ahead of themselves, due to the vehicle in front blocking the line of sight between the driver and the vehicles further ahead. Thus a summation is not used for the instance where the driver is human.

Additionally, an asterisk superscript denotes average (v^* is average velocity). h_i denotes headway (bumper to bumper distance between vehicle i and the next vehicle), x_i denotes position of vehicle i , and l_i denotes the length of vehicle i , such that:

$$h_i = x_{i+1} - x_i - l_i$$

2.2 Simulation Method and Results

As the governing equation works based on a given headway, to make a simulation of this model, we needed to slightly alter the equations to include the dimension of time. To do this, we included a `TIMESTEP`, denoted by δ_t . The vehicles calculate their accelerations every δ_t . This number can be set at an arbitrarily small value which balances the computational intensity and the accuracy lost by having a large timestep. As `f(a)` outputs an acceleration, we used Euler's Method in order to determine velocities and positions of the vehicles.

We were able to change parameters used within the governing equation (such as α_h and β_h), observing the effects they had on the stability of the system. The ring road model allowed the simulation of larger-scale dynamics (such as traffic waves) with the use of fewer vehicles. We also varied other design parameters of the system, as it was possible to change track length, number of vehicles, reaction times, acceleration saturation, and maximum velocity of each of the vehicles individually.

Table 1 shows the parameter values used in the simulation which we have found to be most stable, which are based on parameter values in (1). We changed these parameters and observed how they affected the stability of the system.

The scenario explored in (1) was 3 vehicles (two human, one autonomous) on a circular ring road. In this case, the vehicles are further than h_{go} apart from one another, and therefore accelerate fastest, reaching their maximum velocity v_{max} without interfering with the velocities of any adjacent vehicles. Figure 2 plots the velocities of these three vehicles over time, showing that they stabilise at v_{max} within 20 seconds, with minimal oscillations of velocity.

Total vehicle no. N	3	Max speed v_{max}	$30m/s$
CCC vehicle no. M	1	Stop dist. h_{stop}	$5m$
Human delay τ	$1s$	Driving dist. h_{go}	$100m$
CCC delay σ	$0.2s$	Lower satur. limit a_{min}	$-6m/s^2$
Human gain α_h	$0.2\ 1/s$	Upper satur. limit a_{max}	$3m/s^2$
Human gain β_h	$0.2\ 1/s$	Smoothness range c	$0.05\ m/s^2$
Timestep δ_t	$0.02\ s$		

Table 1: Parameters initially used in the model.

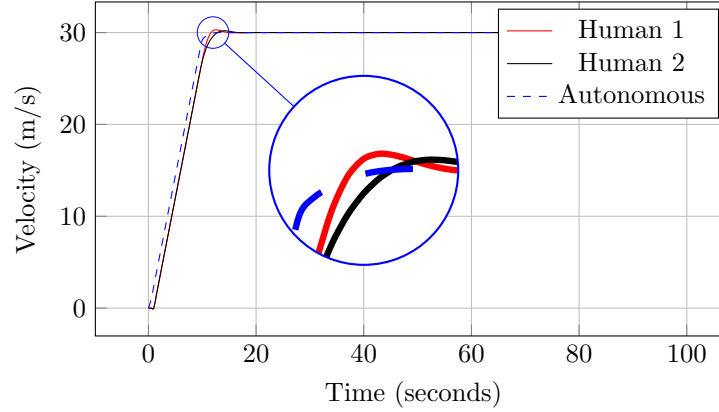


Figure 2: Three vehicles (one autonomous) on an arbitrarily long track. Here the vehicles are spaced apart enough to not affect the other vehicles' velocities. (Parameters: $\alpha = 2$, $\alpha_h = 0.4$, $\beta = 0.2$ and $\beta_h = 0.2$)

Note that in the magnified section of Figure 2, it can be seen that the human vehicles accelerate too much, speeding to above their v_{max} , before decelerating back to v_{max} . This is due to their larger reaction time in comparison to autonomous vehicles. Conversely, it can be seen that autonomous vehicles do not over-accelerate and smoothly reach the maximum velocity. This is important when traffic waves are considered, since the over/under acceleration of human-driven vehicles causes small velocity changes to amplify and create traffic waves, whereas this does not occur with autonomous vehicles.

Traffic waves are formed when one vehicle makes a small acceleration/deceleration. When in a vehicle-dense environment, this causes the vehicle behind it to react by also making a small acceleration/deceleration. However, since human-driven vehicles have long reaction times, they over-accelerate/decelerate (as seen in Figure 2). This causes the initial small acceleration/deceleration to 'amplify' over a number of vehicles, which causes large oscillations in velocity. Since

autonomous vehicles have V2V communications and lower reaction times than human-driven vehicles, autonomous vehicles do not cause the velocity oscillations to ‘amplify’. Traffic waves are therefore characterised by oscillations in velocity of vehicles. Traffic waves travel opposite to the direction of traffic flow due to the over-deceleration of vehicles causing vehicles to slow down earlier than they need to, effectively ‘moving’ the traffic wave backwards to the flow of traffic.

We can explore the effects that adding autonomous vehicles have on the behaviour of the connected vehicle system by making the track shorter and/or moving the vehicles to with h_{go} of each other, as this causes the vehicles to interact, and change their optimum velocity based on headway to the next vehicle. The control parameters of the autonomous vehicles, α and β , are kept constant and at values of 2 and 0.2 respectively. The test environment consists of five human-driven vehicles on a circular loop of length 300 metres, and the velocity-time graph for all five vehicles is plotted in Figure 3.

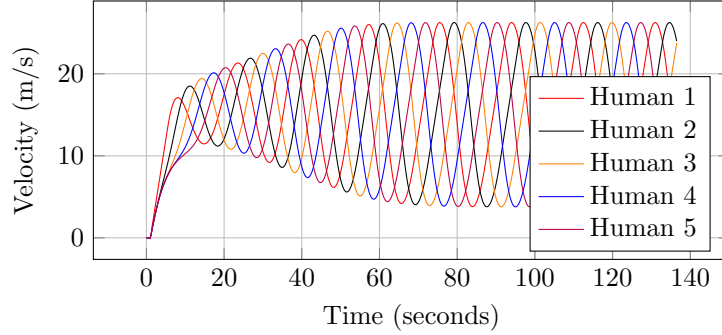


Figure 3: Parameters: $\alpha_h = 0.2$ and $\beta_h = 0.2$

It is observed that, in Figure 3, the velocities of all vehicles are oscillating and not reaching stability. By replacing two of the human-driven vehicles with autonomous vehicles in the same position, 2 cascades are created.

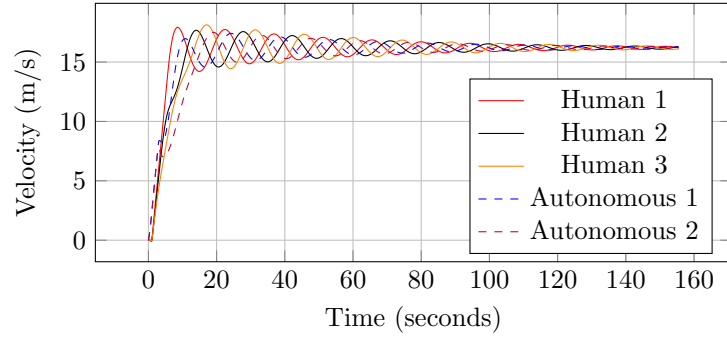


Figure 4: Parameters: $\alpha_h = 0.2$ and $\beta_h = 0.2$

Figures 3 and 4 demonstrate that the addition of autonomous vehicle and V2V communication to a road system can eliminate the previously-seen velocity oscillations and keep the traffic flow smooth. These results agree with stability analysis results obtained in (1, 4), and have been experimentally supported (5). In order to vary behaviours of vehicles in the simulation, the control parameters α , α_h , β and β_h are used, where h stands for human. From Equations 1 and 2, it can be seen that changing the relevant autonomous parameters would yield a similar set of results as changing human parameters. Figures 5 and 6 can be compared to Figure 4 in order to analyse the effects of changing the human control parameters.

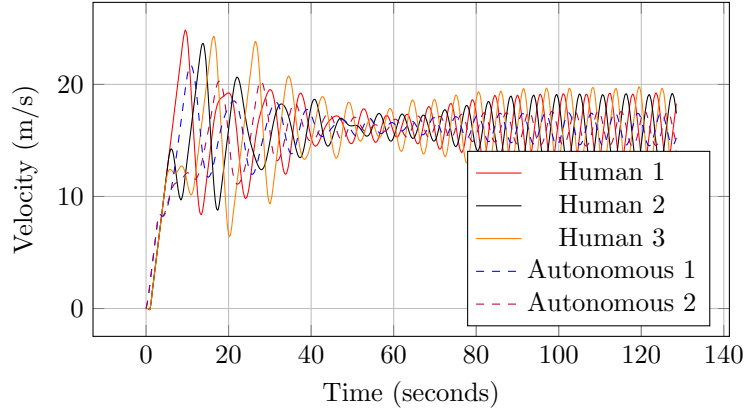


Figure 5: Parameters: $\alpha_h = 0.8$ and $\beta_h = 0.2$

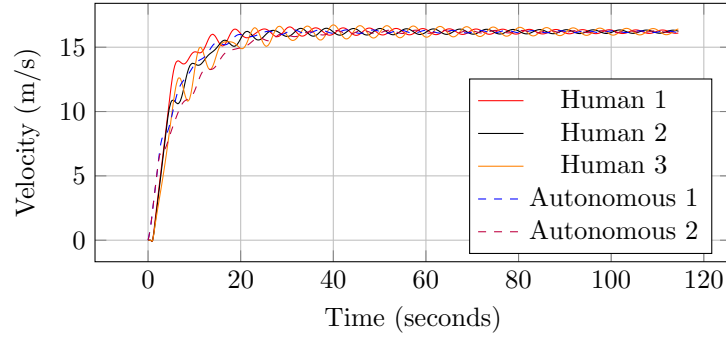


Figure 6: Parameters: $\alpha_h = 0.2$ and $\beta_h = 0.8$

These results clearly illustrate high values of α_h cause the vehicles to get to their optimum velocity quickly, which causes larger oscillations in the velocities of the vehicles. However, this is an unrealistic model and the most suitable value for α_h has been found to be around 0.2 - 0.4, which balances both the stability of the system and real-world data collected in (6).

Traffic waves can be simulated by placing many more vehicles on a longer track. Here, a track of length 720 metres is created with nine human-driven vehicles. (The control parameters of the vehicles are: $\alpha = 2$, $\beta = 0.2$, $\alpha_h = 0.2$ and $\beta_h = 0.2$.) It is observed in Figure 7 that the vehicles do not reach a stability, and their velocities continually oscillate - this is a traffic wave. Where one vehicle initially decelerates by a small amount, the subsequent human-driven vehicles decelerate by increasing and amplifying amounts. Along with the human reaction time, this causes the vehicles to form a traffic wave which does not ever stabilise.

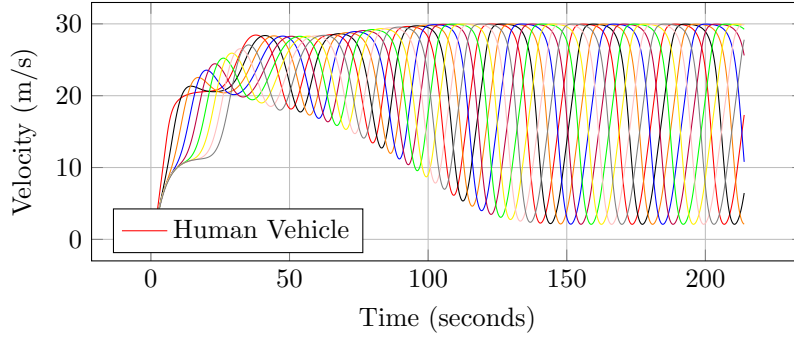


Figure 7: The formation of a traffic wave with nine human-driven vehicles on a track of length 720 metres.

By replacing three of the nine human-driven vehicles with autonomous vehicles, cascades of human-driven vehicles are created, causing the system to reach a

stability where no traffic wave is formed.

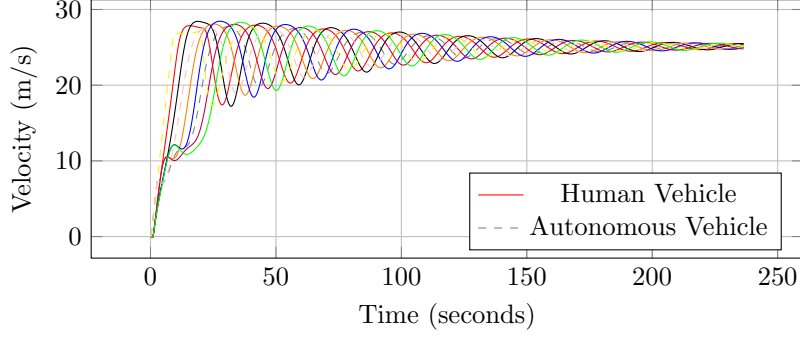


Figure 8: The replacement of some human vehicles by autonomous vehicles eliminates the traffic wave shown in Figure 7.

Figures 7 and 8 show that autonomous vehicles being added in place of some human-driven vehicles have the ability to eliminate traffic waves. This is because the autonomous vehicles form cascades containing a two human-driven vehicles each (cascade length three), reducing the amplifying effects of over-deceleration, reducing the amplitude of the velocity oscillations, reducing the intensity of the traffic waves. Figure 8 shows that the vehicles stabilise after a time of over 250 seconds, converging at a velocity of ≈ 25.14 m/s, which is very close to the value that the optimal velocity function outputs for the system's average headway h^* , showing that the vehicles do eventually reach a stable state (the headways and velocities of all vehicles become stable).

In order to investigate how many autonomous vehicles are needed to mitigate a traffic wave, we created an environment of 20 equally-spaced human-driven vehicles on a track of length 650 metres. (The control parameters of the vehicles are: $\alpha = 2$, $\beta = 0.2$, $\alpha_h = 0.4$ and $\beta_h = 0.2$) This produced a traffic wave, as shown in Figure 9 by the large oscillations in vehicle velocities.

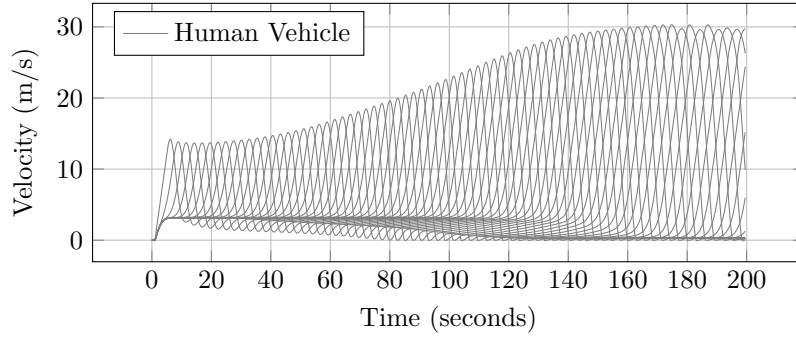


Figure 9: The formation of a traffic wave with 20 equally-spaced human-driven vehicles on a track of length 650 metres.

The replacement of one of the human-driven vehicles with an autonomous vehicle reduces the amplitude of the velocity oscillations (corresponding to a reduction in the intensity of the traffic wave), as shown in Figure 10.

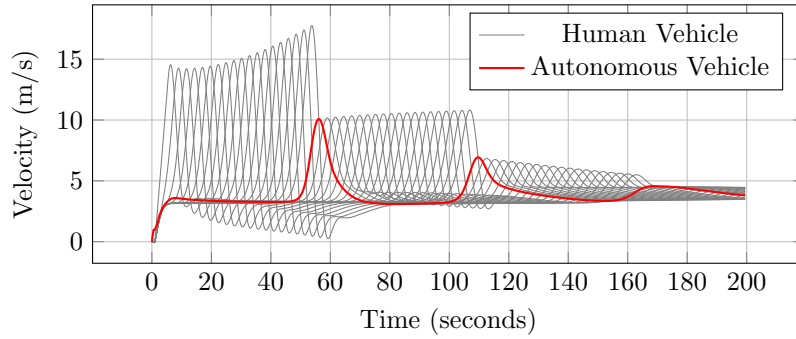


Figure 10: The replacement of one human vehicle with an autonomous vehicle eliminates the traffic wave shown in Figure 9.

The addition of one autonomous vehicle to the system of 20 human-driven vehicles allowed the amplifying effects of humans over-accelerating to be reduced. When the traffic wave reaches the autonomous vehicle, it smoothly accelerates, breaking the feedback loop. If the amplitude decrease by this effect is larger than the increase in amplitude by the compounding over-acceleration, the traffic wave can be eventually eliminated. This example suggests that even if there are only 5% of automatic vehicles in the system, traffic waves can potentially be mitigated. According to (7), this could be achieved optimistically by 2030 and pessimistically by 2040 in some countries.

By adding road features such as traffic lights and variable speed limits, it is possible to create more non-linearities in the system and observe how these affect

the stability of a system.

In the following test scenario, three vehicles (one autonomous, two human-driven) are placed on a track of length 1000 metres, where the main speed limit is v_{max} (30 m/s). Additionally, a 300 metre-long speed limit section of 10 m/s is introduced on the track. Due to limitations in the model, parameter values for human vehicles had to be altered so that vehicles follow the speed limit. Therefore, for test cases involving speed limits, values for α and α_h must be increased, so that the velocity delta does not cause the vehicles to over-speed. The initial results are shown in Figure 11.

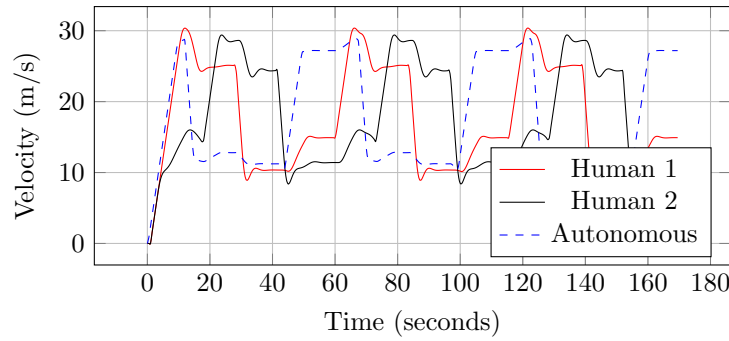


Figure 11: Two human-driven and one autonomous vehicle were placed on track of 1000 metre length with a 300-metre long speed limit section (limit: 10 m/s).

Figure 11 demonstrates that, by making small modifications in the parameter values, it is possible to effectively simulate a scenario with speed limits on the road.

By increasing the number of vehicles on the track, it is possible to simulate a traffic jam (due to a vehicle-dense environment being created). This is represented as a velocity-time graph where the velocities oscillate from zero to a certain velocity - this is shown in Figure 7.

In the following test scenario, we add 3 equally-spaced, 200 metre-long speed limits to the track. Here, these speed limits reduce the velocity of the vehicles to 20 m/s (instead of the standard 30 m/s). Figure 12 shows that the speed limits have the ability to reduce the amplitude of velocity oscillations, corresponding to a decrease in traffic. It can be seen that, at 140 seconds, the vehicles almost reach a constant lower velocity, however the vehicles will never reach a stability due to the additional nonlinearities brought about due to the speed limits. Since speed limits cause vehicles to brake and accelerate, in a vehicle-dense environment, the accelerations of the human-driven vehicles amplify, leading to traffic waves forming. Therefore, the vehicles do not reach a stability through the use of speed limits.

The velocity oscillations have been reduced through the addition of speed limits. However, in order to entirely eliminate the traffic waves, a variable speed limit system (8) would need to be used, however this has not been investigated in this paper.

Autonomous vehicles have a greater effect on eliminating traffic waves than the addition of speed limits to the road, as they are able to create cascades that move smoothly, reducing the amplifying effects of human-driven vehicles over-accelerating - this is demonstrated in Figure 13. In this example, there are no speed limits on the road.

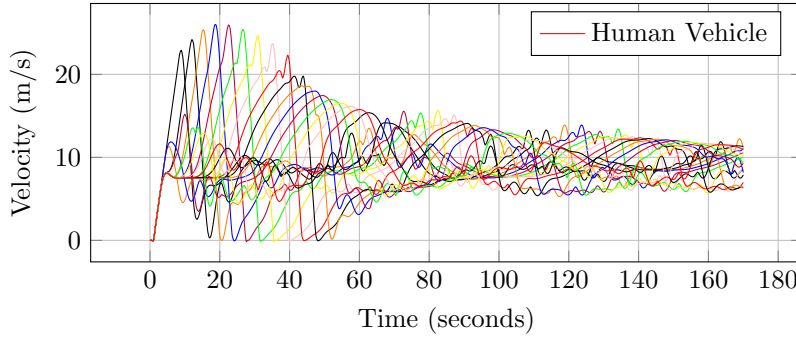


Figure 12: The addition of 3 speed limits to the vehicle-dense environment has the ability to dramatically reduce traffic buildup, however does not have the ability to entirely mitigate traffic.

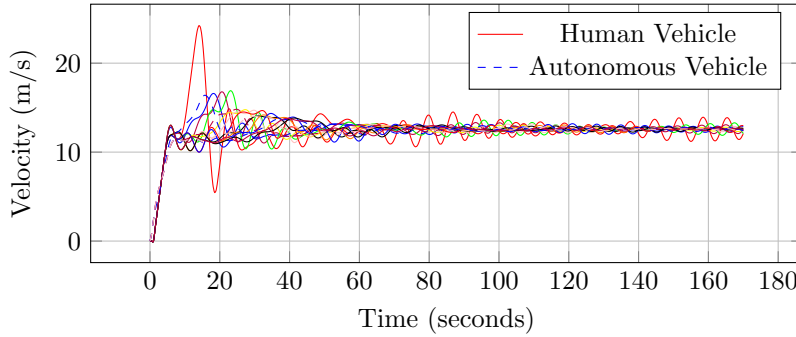


Figure 13: The replacement of every 4th human-driven vehicle with an autonomous vehicle has the ability to entirely eliminate the traffic wave, in an environment without any speed limits, albeit at a lower velocity of ≈ 13 m/s.

Our simulation results obtained agree with the experimental tests in (6). The experimental data was collected using real vehicles, and the full method is described in Appendix A.

3 Stability Analysis

Utilising the simulation we are able to make some estimates on the stability of non-linear systems by observing changes in the model made when parameters are changed. Key parameters include α_h , β_h , track length and the number of vehicles. There are significant limitations to this method as only some key situations were analysed, but this nonetheless allowed us to observe and make some conclusions about the effects of a heterogeneous CVS in a non-linear system.

Three diagrams are produced in Figure 14. The first diagram (Figure 14.1) represents an environment with no autonomous vehicles (all human-driven vehicles). The second diagram (Figure 14.2) represents an environment with a low percentage of autonomous vehicles (10% autonomous vehicles). The third diagram (Figure 14.3) represents an environment with a high percentage of autonomous vehicles (25% autonomous vehicles).

Each square in these diagrams represents a separate situation where only one parameter is changed from the default settings. The defaults used were: 60 cars, track length 3250 metres, α_h 0.2, β_h 0.2, α 1, β 0.2. When run on its own, this system is robustly unstable. The red regions represent where the system is robustly unstable, and the black regions are where a crash occurred. Colours on the gradient between white and blue represent the time taken for the system to reach stability, where a more blue colour refers to a longer time taken to reach robust stability, and where a white region represents robust stability being reached immediately.

Each situation was simulated multiple times with a random offset applied to each vehicle from an equally spaced position to introduce non-linearities. A crash is recorded when at least one of these situations resulted in one.

Note that all bistable situations were also marked as unstable. This is a limitation in this representation method as all instabilities are not equal. Some of these instabilities resulted in a different number of traffic waves, a traffic wave with increasing amplitude, or a traffic wave that did not continue to increase in amplitude. We will discuss this where significant in our analysis.

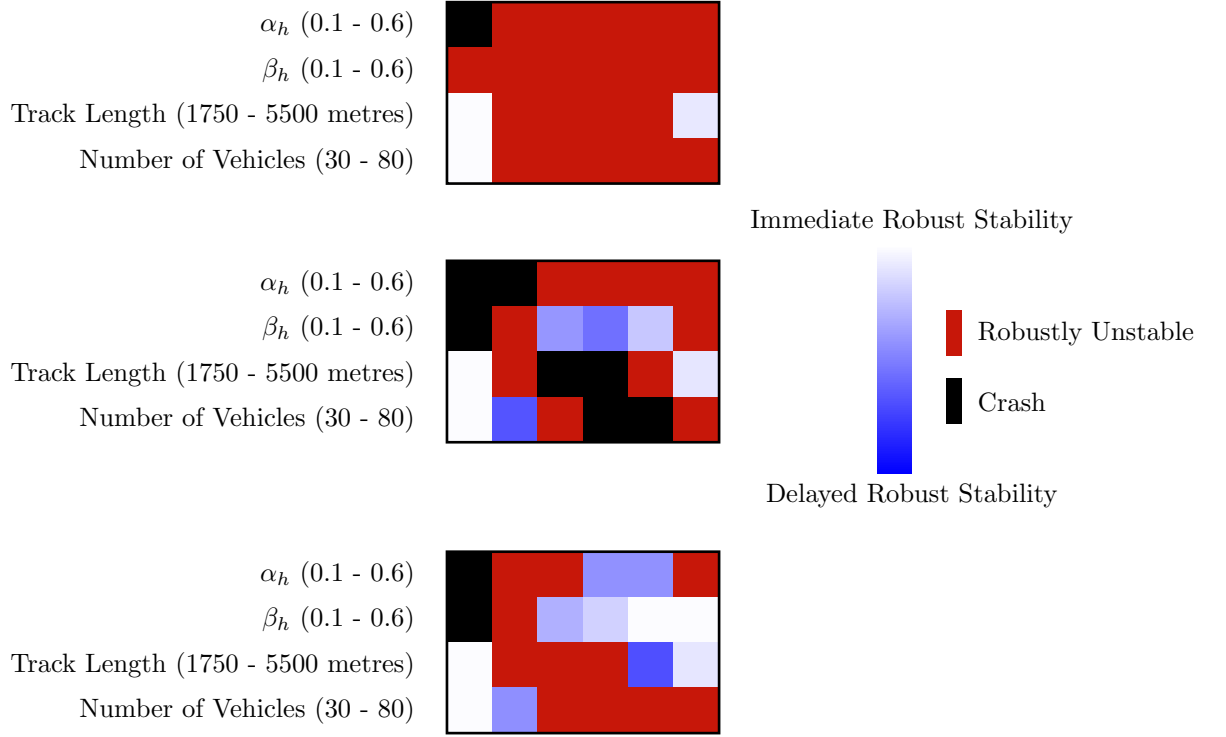


Figure 14: Discrete stability analysis performed on scenarios with varying parameters, showing how the percentage of autonomous vehicles present affects the stability of the system (when varying parameters).

A higher α_h of 0.4 - 0.5 allows the system to stabilise with a high percentage of autonomous vehicles, but for all other systems this system is unstable. When we observed the results, we found that the higher α_h causes more over-accelerations to happen in a shorter period, creating more traffic waves in systems with no or low percentage of autonomous vehicles. Conversely, in a system with a high percentage of autonomous vehicles, the higher α_h allows the autonomous vehicles to quickly dampen the traffic waves and achieve stability in the system. We can expect this effect to become more prevalent as we increase the percentage of autonomous cars. Note that with a low percentage of autonomous cars, the attempt of autonomous cars to mitigate the traffic wave could sometimes result in more crashes when α_h was too low.

A higher β_h has a noticeably large effect on the stability of the system in heterogeneous systems. The higher β_h allows oscillations within each cascade to be dampened which results in significantly more stable systems. This is also what causes the crashes seen when β_h is too low for both heterogeneous systems. In (6), the realistic value of β_h currently was taken to be 0.4-0.6. This works

well with heterogeneous systems according to our model.

The track length is a significant factor in the stability of the system. For this system, when the track length is around 1750m, the density of cars is high and the velocity so low that the over-accelerations no longer result in a traffic jam. When the track length is around 5500m, $h^* \approx h_{go}$, meaning that all vehicles can go at the speed limit without risking a crash. Significantly, we see that the system can crash when the track length is 3250-4000m with a low percentage of autonomous vehicles. The issue with traffic waves within the cascade is the most prevalent in this region which creates instability. Note that these results also had a high variance, where some initialisations allowed the system to even stabilise. With a high percentage of autonomous vehicles, there were reliably no crashes within this region and also reached stability with a higher traffic density.

A higher number of vehicles causes problems with the low percentage of autonomous vehicles for a similar reason. A higher number density of cars causes more over-accelerations to occur in a shorter length of track, causing a higher likelihood of crashes when the cascades are too long. Once again we see this problem is eliminated with a higher percentage of autonomous vehicles.

4 Programme of Work

After our first meeting with our mentor, we worked towards understanding the governing equations (Equations 1 & 2). We developed an intuitive understanding of these equations as demonstrated in the paper. We then worked on programming a simulation of 3 vehicles on a circular road, with no obstacles, in Python. The position of the vehicles is stored as an angle around the circular road, ϕ . When animating the result, we use simple parametric equations to map the angle to the position on the road. When the length of the road is changed, ϕ now represents the proportion of the road that the vehicle has driven and is scaled appropriately to display correctly. We again faced some difficulty computationally when making these animations due to the large amount of data, especially when trying to decrease δ_t , so the animation only renders every n frames, where n is arbitrarily decided.

At first, we started by modeling the situation with only human-driven vehicles. Only after succeeding at this did we implement autonomous vehicles. Through the use of object-oriented programming, we found that we were able to individually change the parameters of each of the vehicles, which allowed us to experiment with an incredibly wide range of parameters. This also allowed us coincidentally to put as many vehicles as we would like on the track. We plotted the results using Matplotlib, plotting acceleration, velocity and headway for each vehicle, representing each vehicle with a separate line on a line graph. Through experimentation, we found interesting results, after which we recorded the parameters that caused said result and investigated it further, in particular to try and find examples of stability boundaries. Without the introduction of

reaction times, denoted in the equation as θ we found that all systems that we observed always reach a robust stability. After implementing reaction time, done by keeping a history of headways and utilizing a headway from time θ prior, we started observing interesting real-life situations such as traffic waves and vehicle crashes. We found that without a velocity delta, one deceleration in a vehicle at the front of the system would transfer through the system, amplifying due to each drivers' reaction times causing a crash. This effect can still be seen in the final simulation with a small β_h and a high speed limit. In some situations we observed bistability, a cycle of accelerations and decelerations that never stabilise but continue endlessly. Furthermore multiple orders of these cycles could sometimes be observed, although we could not mathematically confirm this. In (1), Kiss et al. write that the boundaries can be represented by a Hopf bifurcation, however, we were not able to replicate this due to limitations in our understanding.

We then added standard road features such as traffic lights and variable speed limits. For traffic lights, when the traffic light is green, it is 'invisible' to all other vehicles in the sense that it does not affect the headway and therefore the OV. When it is orange, the vehicle closest behind the traffic light will judge if it can safely stop at the traffic light in time. We use a simple constant acceleration equation plus a buffer to simulate this decision. For vehicles that decide that they are able to pass the traffic light, the traffic light remains 'invisible'. For vehicles that have decided that they must stop, the traffic light is 'visible' and acts like another vehicle in the sense that the distance to the traffic light will be interpreted as it were a headway. We thought about how the velocity delta makes little sense in this scenario and experimented with attempts to remove it, but the results obtained were largely unsuccessful. A large limitation in our method that should be mentioned here is that human vehicles should be able to see over other vehicles and see the traffic light. In our simulation, since humans only look up to the next vehicle, due to the sudden change in headway caused by the change of colour the vehicle in front suddenly brakes, after which the vehicle behind is more likely than it is in reality to crash. Additionally, this model is flawed in that, vehicles do not know how long the traffic light has been orange for - in reality, humans are able to judge roughly how long the traffic light has been orange for, and then make a decision on whether they can pass the traffic light or not. This does not occur in this model, and therefore some vehicles pass the traffic light at the exact same moment that the traffic light turns from orange to red (therefore registering as a crash). This made it difficult to gather data for traffic lights, as many erroneous crashes were detected due to this issue.

Regarding speed limits, drivers adjust to the new speed limit when they are within h_{go} of the new speed limit; effectively when they 'see' the speed limit. The new speed limit does not change the shape of the OV function and still smoothly accelerates and decelerates as expected.

We then worked to solve the linear stability analysis equations, which would

allow us to analyse which parameter values would lead to a stable connected vehicle system. By using the MDBM method (9) we were able to solve the linear stability analysis, producing graphs of stability in parameter space. This was done computationally, and we wrote the program to solve the linear stability analysis in Python. This allowed us to mathematically analyse the effects that changing parameter values would have on the vehicles. The equation itself is easy to evaluate for any complex number simply by using NumPy. We ran into an issue in terms of computational power when we tried to make the mesh sufficiently dense to find all characteristic roots. We had many issues with what looked like false roots and some roots that seemed to exist not appearing, but since we strongly observed the same circular ‘droplets’ seen in (1), we were confident that we were finding some good solutions. A limitation of this method is that we would not be able to find stability boundaries for our more complex models, and therefore we did not pursue this method further. However, inspired by other papers that we read, we realised that we could investigate some key scenarios and determine the effect of the design parameters on these systems. We discretely changed variables that we were interested in, while making sure to keep within realistic values obtained from (6).

5 Conclusion

In this paper, the role of CAVs within heterogeneous systems of vehicles was explored and yielded results which showed that they effectively increase the flow of traffic. Many aspects such as the delay between the mathematical calculations of the governing equations used for the automated vehicle and the breaks being applied, and the human reaction speed, were taken into account to ensure the model is as realistic as possible. The governing equations described linear systems, but non-linearity was also explored through traffic lights and variable speed limits, to which the CAVs effectively decreased the risk of traffic waves emerging and causing congestion. A discrete linear stability analysis was conducted by running the program with different parameters and plotting stable configurations on a parameter-space diagram.

Glossary

Acceleration Saturation - the acceleration capabilities of a vehicle.

Cascade - a ‘block’ of vehicles from an autonomous vehicle to the next automated vehicle, but not including it. They move smoothly in a group.

Headway - the distance between two vehicles (bumper to bumper).

Heterogeneous - a system that contains a mixture of human and autonomously driven vehicles.

Stability Analysis - when a system reaches a state where the headways are

equal (in the context of CVS). Consequentially, all velocities are equal and acceleration is zero.

Vehicle to Vehicle (V2V) Communication - the transfer of information (especially velocity) to an automated vehicle from all vehicles up to and including the next automated vehicle.

A Parameters and Real-World Data

In (6), three vehicles are placed on a straight, single-lane public road, with the autonomous vehicle behind two human-driven vehicles. This forms one cascade of vehicles.

A straight, single-lane road was chosen over a circular track in order to minimise lateral acceleration and the need for the vehicle to constantly be turning - this allowed for experimental speeds to reach the high simulations speeds, whilst keeping lateral acceleration to a comfortable limit. The experiment was set up in a way that the CCC algorithm still acts as if the vehicles were on a ring road.

The experiment recorded the headways, velocities and accelerations of all 3 vehicles over time, from which graphs were plotted for velocity against headway for the two human drivers. This allowed for the parameters used in our model to be calculated, and for simulations of the experimental setup to be made. The parameters that were obtained from this experiment were:

- Stop dist. h_{stop}
- Driving dist. h_{go}
- Max. speed v_{max}

References

- [1] Adam K Kiss, Sergei S Avedisov, Daniel Bachrathy, and Gábor Orosz. On the global dynamics of connected vehicle systems. *Nonlinear Dynamics*, 96: 1865–1877, 2019.
- [2] Gábor Orosz, R Eddie Wilson, and Bernd Krauskopf. Global bifurcation investigation of an optimal velocity traffic model with driver reaction time. *Physical Review E*, 70(2):026207, 2004.
- [3] Masako Bando, Katsuya Hasebe, Akihiro Nakayama, Akihiro Shibata, and Yuki Sugiyama. Dynamical model of traffic congestion and numerical simulation. *Physical review E*, 51(2):1035, 1995.
- [4] Tamás G Molnár, Michael Hopka, Devesh Upadhyay, Michiel Van Nieuwstadt, and Gábor Orosz. Virtual rings on highways: Traffic control by connected automated vehicles. In *AI-enabled Technologies for Autonomous and Connected Vehicles*, pages 441–479. Springer, 2022.
- [5] Raphael E Stern, Shumo Cui, Maria Laura Delle Monache, Rahul Bhadani, Matt Bunting, Miles Churchill, Nathaniel Hamilton, Hannah Pohlmann, Fangyu Wu, Benedetto Piccoli, et al. Dissipation of stop-and-go waves via control of autonomous vehicles: Field experiments. *Transportation Research Part C: Emerging Technologies*, 89:205–221, 2018.
- [6] Sergei S Avedisov, Gaurav Bansal, Adam K Kiss, and Gábor Orosz. Experimental verification platform for connected vehicle networks. In *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*, pages 818–823. IEEE, 2018.
- [7] Ahmadreza Talebian and Sabyasachee Mishra. Predicting the adoption of connected autonomous vehicles: A new approach based on the theory of diffusion of innovations. *Transportation Research Part C: Emerging Technologies*, 95:363–380, 2018.
- [8] Meng Wang, Winnie Daamen, Serge P Hoogendoorn, and Bart Van Arem. Connected variable speed limits control and car-following control with vehicle-infrastructure communication to resolve stop-and-go waves. *Journal of Intelligent Transportation Systems*, 20(6):559–572, 2016.
- [9] Dániel Bachrathy and Gábor Stépán. Bisection method in higher dimensions and the efficiency number. *Periodica Polytechnica Mechanical Engineering*, 56(2):81–86, 2012.